

What is internet protocol?

The internet protocol or IP is what makes the internet possible. It is responsible for routing and addressing packets of data so that they can travel across networks and arrive at the correct destination. In this layer, data is divided into packets and also reassembled for incoming packets.

IP is a connectionless service once the data reaches the destination, but it acts as per the transport layer protocol, usually TCP (connection-oriented) or UDP (connectionless), commonly called TCP/IP or UDP/IP, respectively.

What is an IP packet?

IP packets are created by adding an IP header to each packet of data before it is sent on its way. Each packet contains two parts — the header and the payload. A payload is the data that is transmitted. An IP header records several pieces of information about the packet, such as:

- Source IP address
- Destination IP address
- Header length
- Packet length
- Time to live (TTL), or the number of network hops a packet can make before it is discarded
- Checksum, for error checking in the received data
- Transport protocol being used (TCP, UDP)

There are 14 fields of information in IPv4 headers, although one of them is optional. The method of adding a header to the payload is called encapsulation.

How does IP routing work?

IP routing is a process to determine the best path for data to transfer from source to destination. The data packets follow the path based on the routing table configuration along with routing algorithms. The routing algorithm considers various factors like the size of the packet and its header to determine the most efficient route for the data from its source to the destination. Packets travel from the router following the routing table that indicates the next hop's address, until they reach the targeted IP address. A variety of routing protocols help route packets based on their destination IP addresses. A 'traceroute' command helps to understand the route a packet takes to reach the destination.

How do IP addresses work?

An IP address is a unique identifier assigned to the computer that is connected to the internet. IP protocol needs to add the IP address of the source and destination to the header.

The source IP has two types of addresses.

Public IP address: The scope of the public address is global, which means that we can communicate outside the network. This address is assigned by the ISP (internet service provider). It is not available free of cost.

The 'dig +short myip.opendns.com @resolver1.opendns.com' command will show the public address attached to the system. It could also be Googled by asking 'What is my IP?'

Private IP address: The scope of this address is local, as we can communicate within the network only. It is generally used for creating a local area network and is available free of cost. The 'ifconfig' command will show the private IP address attached to the system.

Two types of private IP addresses are assigned.

- **Static:** Here, the assigned IP address does not change but can be changed as part of network administration. It helps to find out the device its mapped to.
- **Dynamic:** In this case, the IP address keeps changing; the dynamic host configuration protocol or DHCP server assigns the IP address. A DHCP server needs to be configured with an appropriate pool of IP addresses, which it will assign to client devices as they join the network, ensuring that the addresses are unique. DHCP is very useful when the network has many host devices and reduces the chance of errors in assigning addresses.

The next question is: what is the source IP address? The packet uses the public IP address of the computer as its source; however, a LAN/WAN uses a private IP address.

We can use a private IP address in our private network for communication. But if the destination is outside the network, and needs to access the internet, one public IP address is needed.

Network address translation (NAT) is a process in which one or more local IP addresses are translated into one or more global IP addresses and vice versa in order to provide internet access to local hosts. NAT masks the port number of the host with another port number in the packet that is routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates in a router or firewall.

Let's now talk about the destination IP address. Assume we want to connect

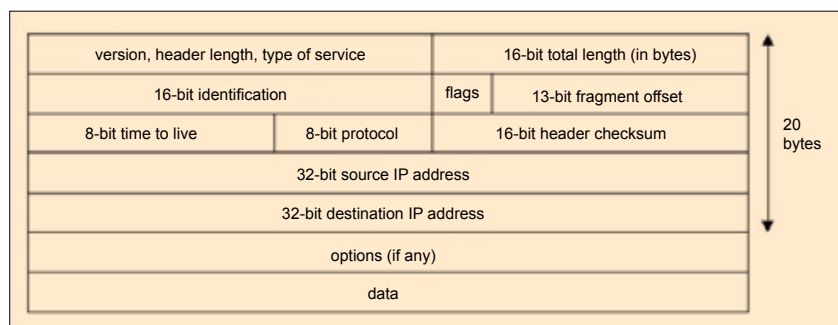


Figure 4: IP packet structure (Source: <https://web.stanford.edu/class/msande91/si/www-spr04/readings/week1/InternetWhitepaper.htm>)